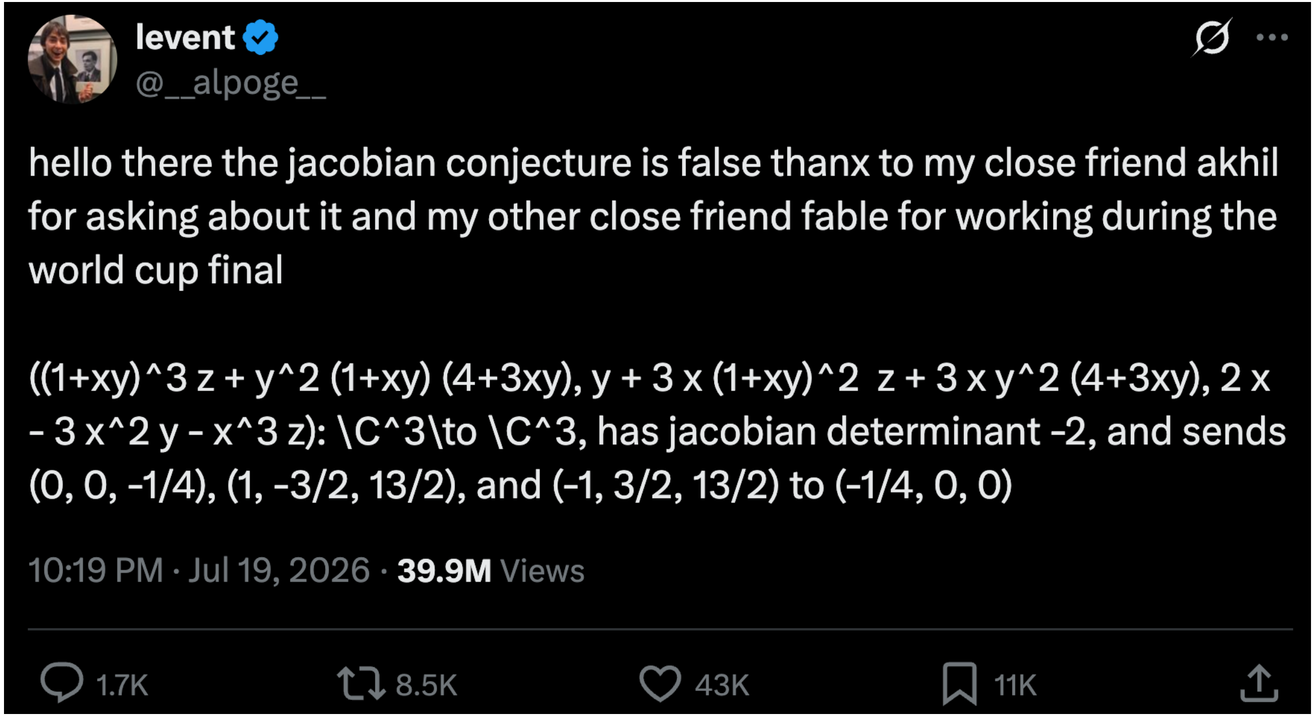
Course Overview

CSCI 601-471/671 (NLP: Self-Supervised Models)

Why This Course?

- Let’s start with an example:



levent
@__alpoge__

hello there the jacobian conjecture is false thanx to my close friend akhil for asking about it and my other close friend fable for working during the world cup final

((1+xy)^3 z + y^2 (1+xy) (4+3xy), y + 3 x (1+xy)^2 z + 3 x y^2 (4+3xy), 2 x - 3 x^2 y - x^3 z): \C^3\to \C^3, has jacobian determinant -2, and sends (0, 0, -1/4), (1, -3/2, 13/2), and (-1, 3/2, 13/2) to (-1/4, 0, 0)

10:19 PM · Jul 19, 2026 · **39.9M** Views

1.7K 8.5K 43K 11K

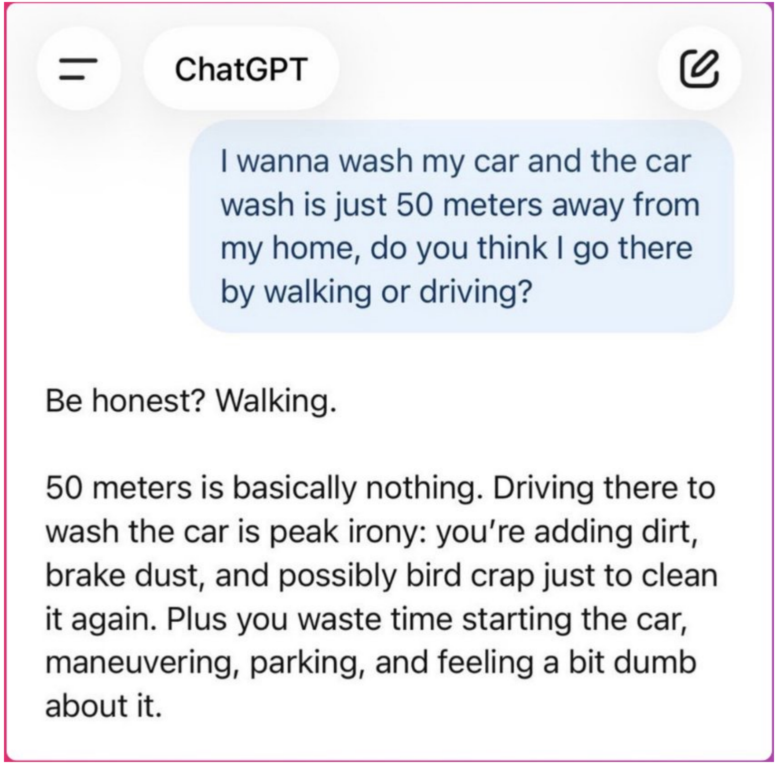
Jacobian conjecture, a disproven conjecture

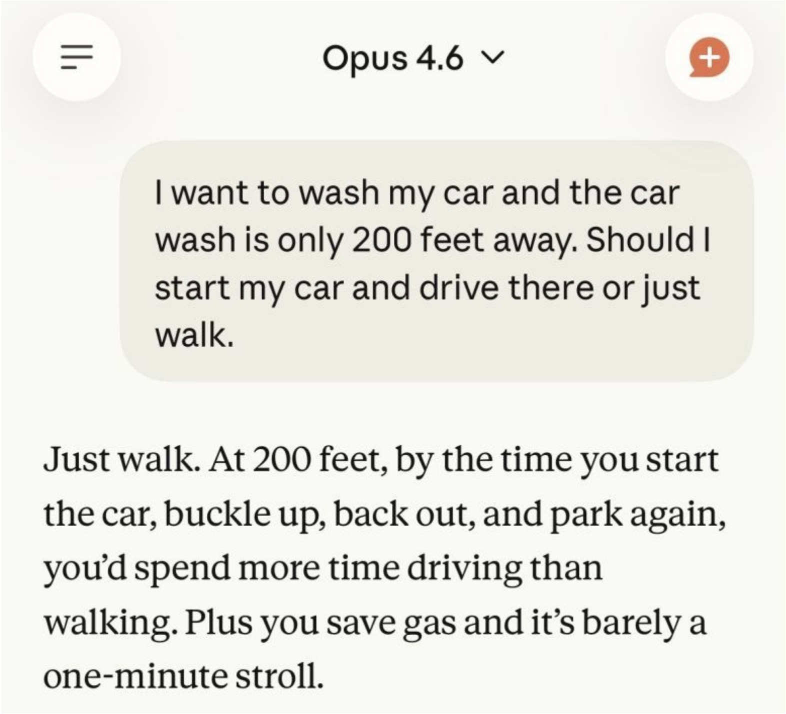
- Very simple-sounding statement about polynomial.

> *if a polynomial map has an everywhere nonzero constant Jacobian determinant, then it should be globally invertible, with a polynomial inverse.*

- Posed by Ott-Heinrich Keller in 1939.
- Remained open for ~90 years.
- Until someone found a counter example for it (for dim>2) with the help of their Claude agent.

Meanwhile …


ChatGPT
I wanna wash my car and the car wash is just 50 meters away from my home, do you think I go there by walking or driving?
Be honest? Walking.
50 meters is basically nothing. Driving there to wash the car is peak irony: you're adding dirt, brake dust, and possibly bird crap just to clean it again. Plus you waste time starting the car, maneuvering, parking, and feeling a bit dumb about it.


Opus 4.6
I want to wash my car and the car wash is only 200 feet away. Should I start my car and drive there or just walk.
Just walk. At 200 feet, by the time you start the car, buckle up, back out, and park again, you'd spend more time driving than walking. Plus you save gas and it's barely a one-minute stroll.

LLMs are simultaneously astonishingly capable and …

What LLMs can do
Write/debug substantial programs
Solve olympiad-level math problems
Operate computers / act as agents
Synthesize enormous literatures

What enables them to do these? 🤔

- What are the principles that govern this technology?
- Can we replicate it?
- Can we extend it?

LLMs are simultaneously astonishingly capable and surprisingly brittle

What LLMs can do	What LLMs still struggle with
Write/debug substantial programs	Fail simple logic puzzles
Solve olympiad-level math problems	Lose track of constraints
Operate computers / act as agents	Hallucinate confidently
Synthesize enormous literatures	Break under small distribution shifts

- Can we explain these?
- Can we predict them?
- Can we mitigate (or even better, solve) them?

Course Learning Objectives

- We will cover a variety of inter-related topics:
  - Architectures
  - Pre-training
  - Alignment and safety
  - Efficiency
  - Interaction with the world (code, physical world, etc.)
  - Impacts on humans -- their misuse, biases, etc.
- Skills:
  - **Technical**—understanding of the algorithms and implementing them.
  - **Soft skills**—intuition about capabilities, teamwork, algorithmic thinking.

Focus on Natural/Human Language

- **Most** of the class revolves around **natural language**.

- Why natural language?
  - It is a **convenient medium of communication**.
  - Natural language is our species' best attempt to encode **everything about the world** as **efficiently** as possible.
  - A huge archive of natural language is **freely available** (e.g., on the web).

Self-Supervised Models

Self-Supervision

[Slide credit: Colin Raffel]

Self-Supervision

[Slide credit: Colin Raffel]

Self-Supervision

[Slide credit: Colin Raffel]

Self-Supervision

Dataset of natural images

[Slide credit: Colin Raffel]

Self-Supervision

Dataset of natural images

Generated image, from "Large Scale GAN Training for High Fidelity Natural Image Synthesis", Brock et al.

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Self-Supervision

== treaty of paris (1763)

the treaty of paris, also known as the treaty of 1763, was signed on 10 february 1763 by the kingdoms of great britain, france and spain, with portugal in agreement, after great britain's victory over france and spain during the seven years' war.

the signing of the treaty formally ended the seven years' war, known as the french and indian war in the north american theatre, ….

[Slide credit: Colin Raffel]

Self-Supervision

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== wheelbarrow

A wheelbarrow is a small hand-propelled vehicle, usually with just one wheel, designed to be pushed and guided by a single person using two handles at the rear, or by a sail to push the ancient wheelbarrow by wind. The term "wheelbarrow" is made of two words: "wheel" and "barrow." "Barrow" is a derivation of the Old English "barew" which was a device used for carrying loads. The wheelbarrow is designed to …..

[Slide credit: Colin Raffel]

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== lemon

WIKIPEDIA
The Free Encyclopedia

The lemon (Citrus limon) is a species of small evergreen trees in the flowering plant family Rutaceae, native to Asia, primarily Northeast India (Assam), Northern Myanmar or China.[2] The tree's ellipsoidal yellow fruit is used for culinary and non-culinary purposes throughout the world, primarily for its juice, which has both culinary and cleaning uses.[2] The pulp and rind are also used in cooking and baking. ….

Dataset of Wikipedia articles

[Slide credit: Colin Raffel]

Sel pervision

WIKIPEDIA
The Free Encyclopedia

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Dataset of Wikipedia articles

== wings over kansas

wings over kansas is the second studio album by jason ammons, john bolster and mo rosato. the album debuted at number one on the billboard 200, selling 35,000 copies in it first week at the time. it was the second highest selling album to debut at the billboard top 50 and the third highest selling album to debut at the top heatseekers, with 26,000 copies sold. this is the supremes album earning the nickname …

[Slide credit: Colin Raffel]

Self-Supervised Models

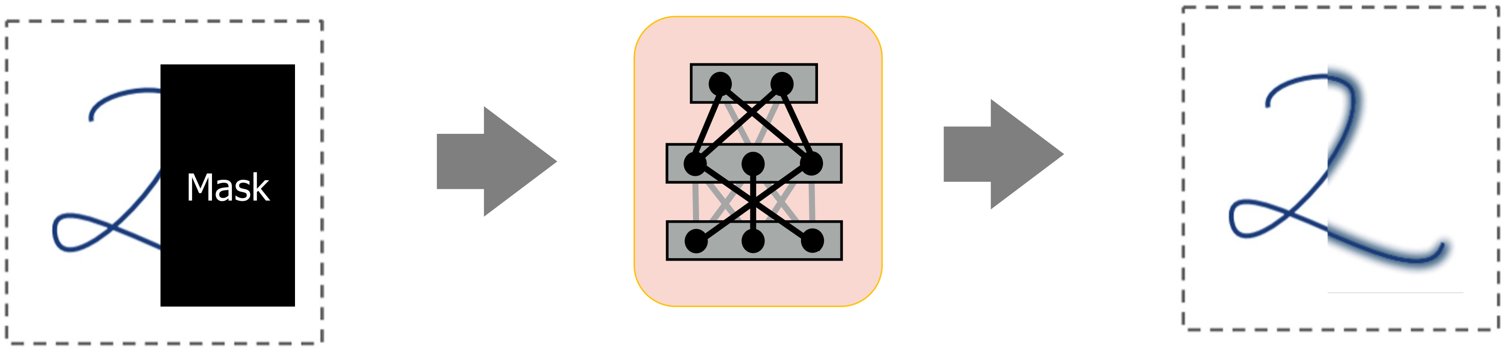
*are **predictive models** of the world!*

22

Self-Supervised Models

*are **predictive models** of the world!*

- Are trained to complete partial samples from the world.



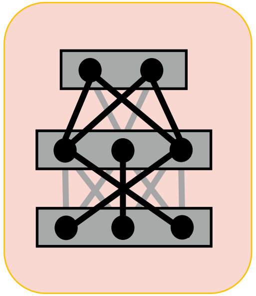
[Bengio et al. 2004, Hinton et al. 2006, Peters et al. 2018, ...]

Self-Supervised **Models**

*are **predictive models** of the world!*

- Are trained to complete partial samples from the world.

“Wings over Kansas is [MASK]”



“Wings over Kansas is an aviation website founded in 1998 by Carl Chance owned by Chance Communications, Inc.”

[Bengio et al. 2004, Hinton et al. 2006, Peters et al. 2018, …]

Self-Supervised Models

*are **predictive models** of the world!*

*learned from **cheaply available** unlabeled data*

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Self-Supervised Models

*are **predictive models** of the world!*

*learned from **cheaply available** unlabeled* ***data***

26

Self-Supervised Models

are tightly connected to tasks we care about.

Here is an example …

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Self-Supervised Models

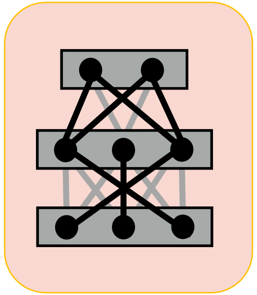
are tightly connected to tasks we care about.

- **Goal:** Answering questions

Question: "Where is the birthplace of the American national anthem?"

✍️

"The birthplace of the American national anthem" [MASK]



"The birthplace of the American national anthem, "The Star-Spangled Banner," lies in Baltimore, Maryland."

Produced with GPT-J: https://6b.eleuther.ai/

Self-Supervised Models

1. *Are predictive models of the world.*
2. *Are learned from unlabeled data.*
3. *Tightly connected to tasks we care about.*

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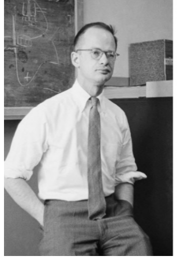
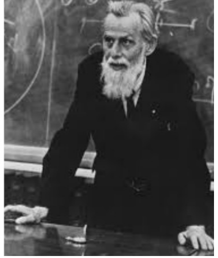
How did we get here?

Progress in AI

- Many advances are due to neural networks
- How old are neural networks?

Progress in AI





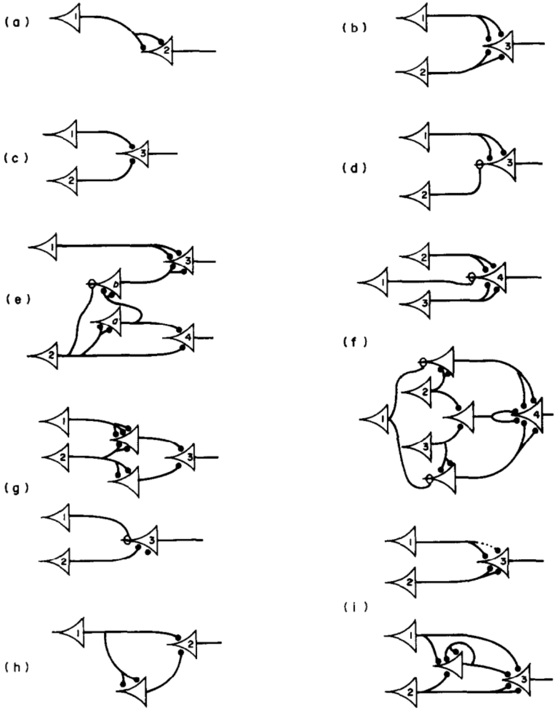
- Many advances are due to neural networks
- How old are neural networks?

McCulloch & Pitts (1943)

A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY*

■ WARREN S. MCCULLOCH AND WALTER PITTS
University of Illinois, College of Medicine,
Department of Psychiatry at the Illinois Neuropsychiatric Institute,
University of Chicago, Chicago, U.S.A.

Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.



[Slide credit: Arman Cohan]

Progress in AI

- Many advances are due to neural networks
- How old are neural networks?
  - They’ve been around since the 1940s
  - But why have only recently we seen breakthroughs?
    - **3 necessary forces had to come together!**

[Slide credit: Arman Cohan]

Force 1: Massive Amount of Data

- Internet provided us with a massive repository of data.

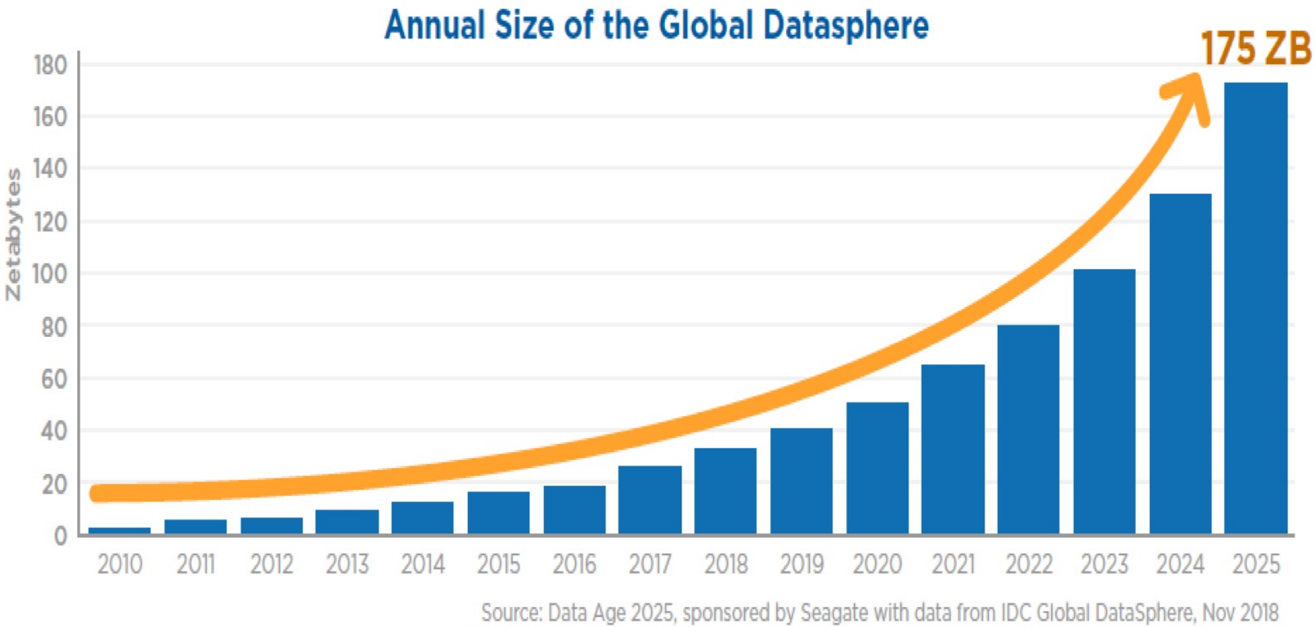
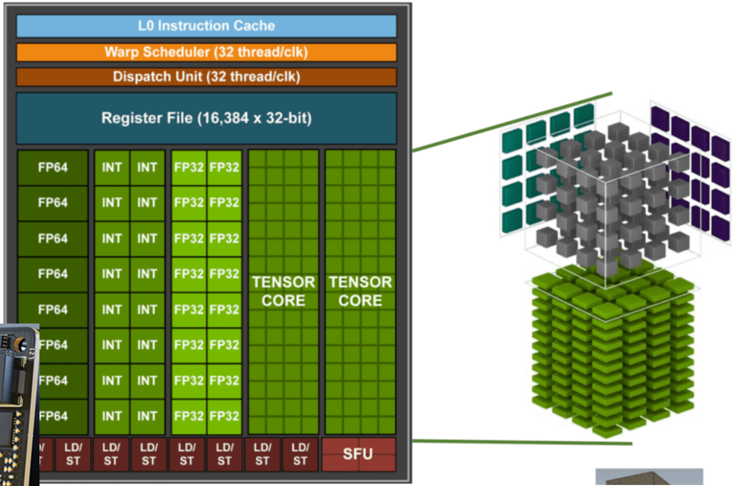
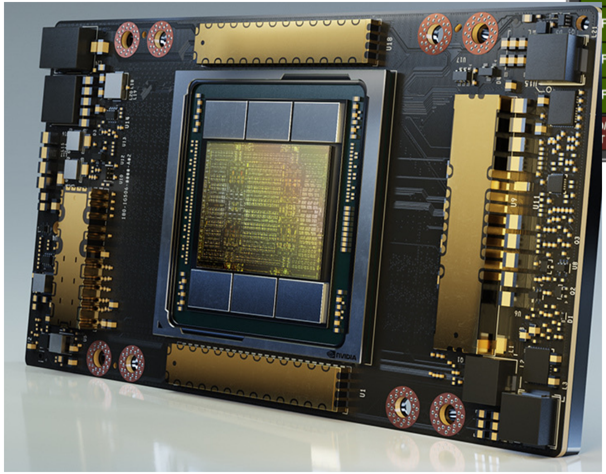


Image source: https://www.forbes.com/sites/tomcoughlin/2018/11/27/175-zettabytes-by-2025/?sh=1e6ca8455459

Force 2: Computing Power

- Fast processors for deep learning!





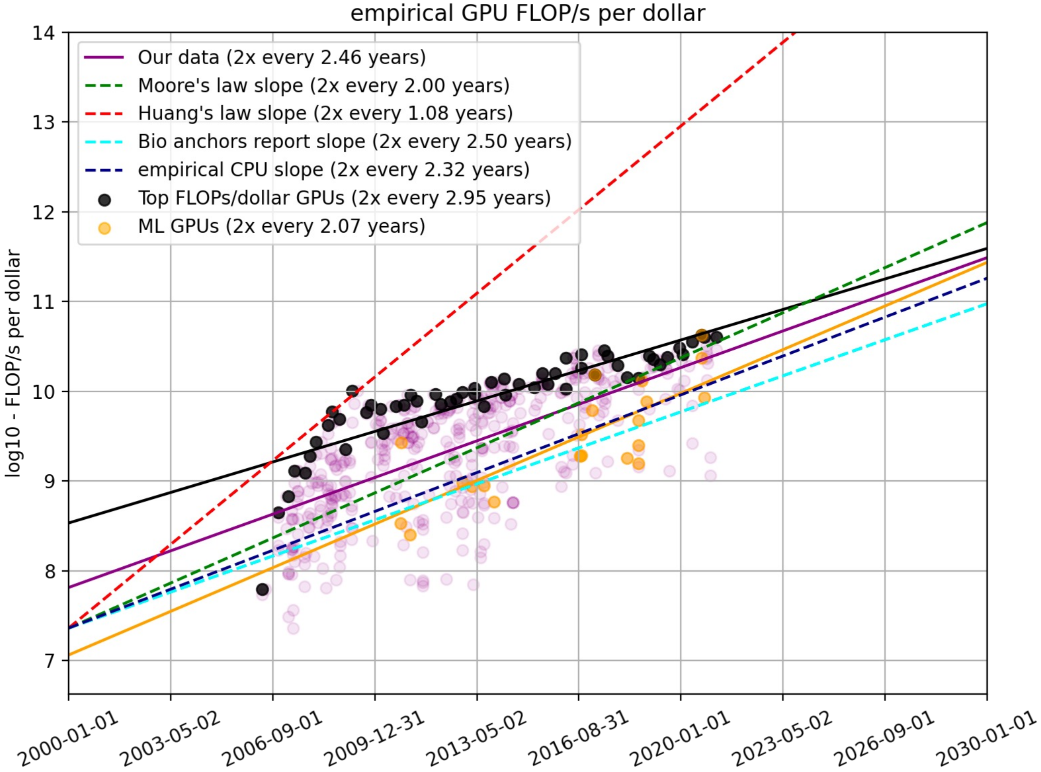
A100 GPU

Force 2: Computing Power

- Fast processors for deep learning!

- Cheaper computing power over time.

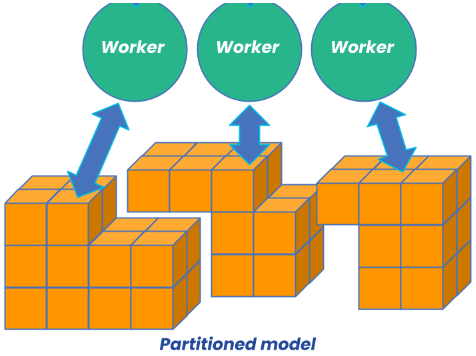
The amount of computing power, per dollar



https://www.lesswrong.com/posts/c6KFvQcZggQKZzxr9/trends-in-gpu-price-performance

Force 2: Computing Power

- Fast processors for deep learning!
- Cheaper computing power over time.
- Distributed training/inference allows us to scale to a larger set of processors.



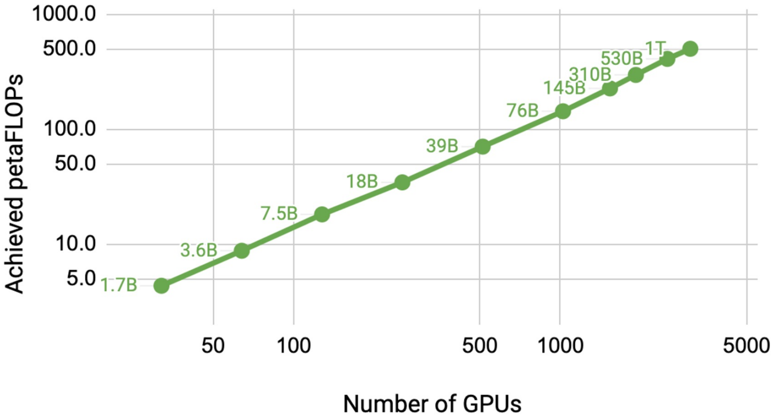
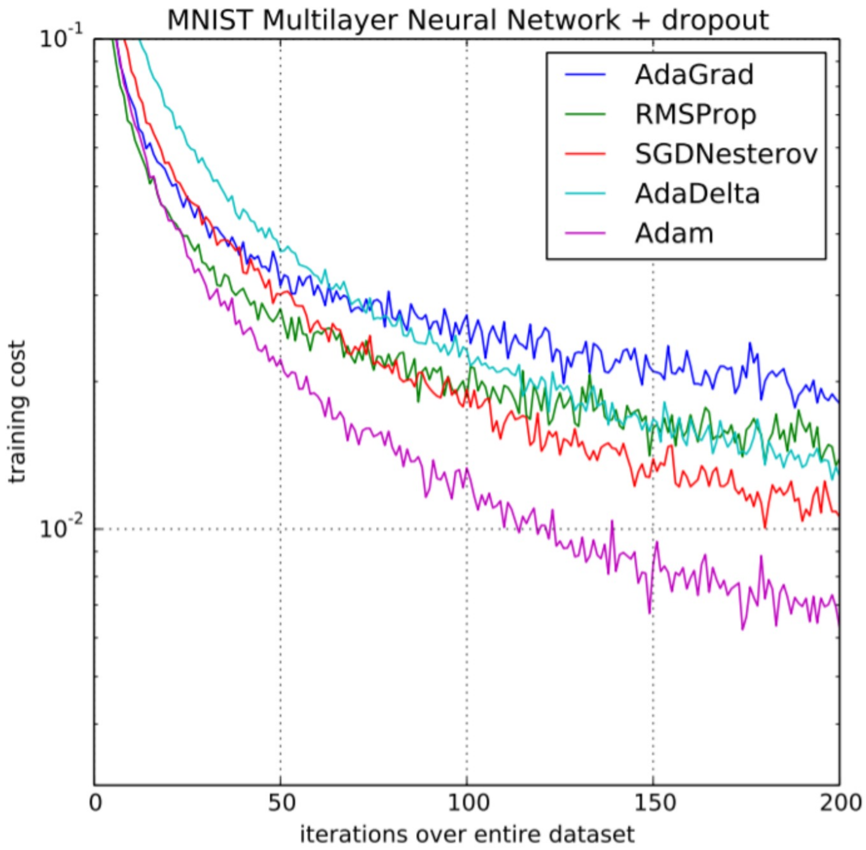


Image source: https://developer.nvidia.com/blog/scaling-language-model-training-to-a-trillion-parameters-using-megatron/

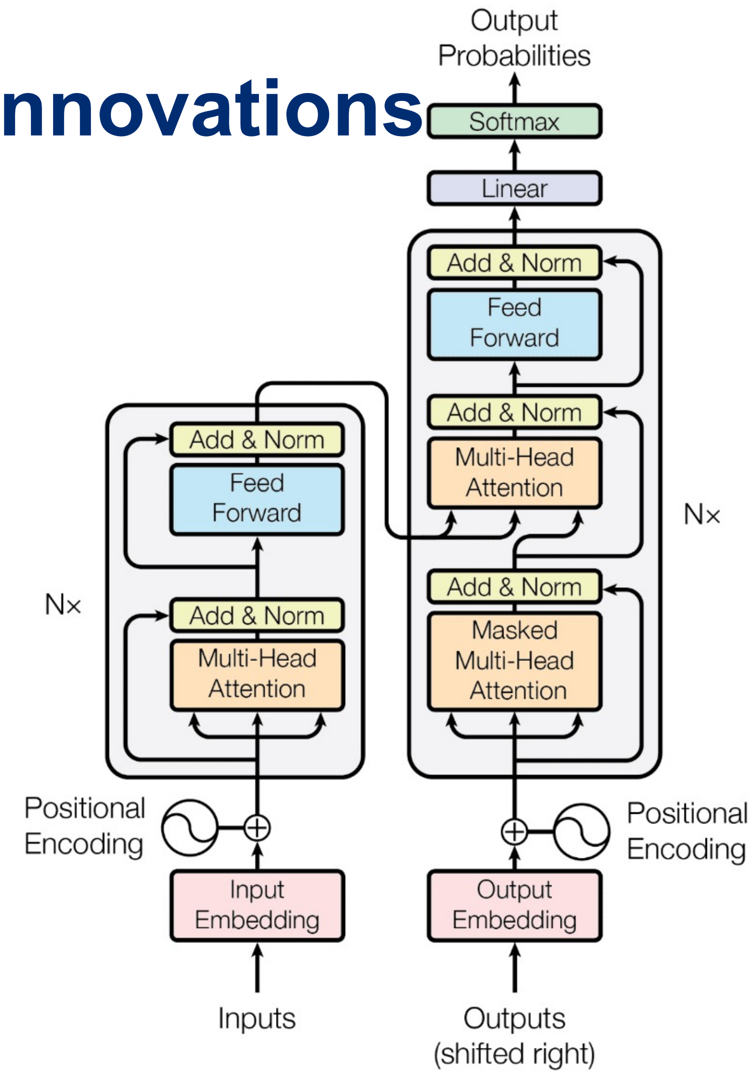
Force 3: Algorithmic innovations

- Advances in optimization



Force 3: Algorithmic innovations

- Advances in optimization
- Innovations in model architectures
- ….



The success we dreamed of

Language models that are remarkably capable at solving many important NLP benchmarks.

Current state of Self-supervised Models

- Almost every AI model is based on Neural networks
- Performance is consistently improving with scale
  - More training data
  - Larger models (number of neural network parameters)

[Slide credit: Arman Cohan]

Current state of Self-supervised Models

State-of-the-art models are
hundreds of billions of parameters

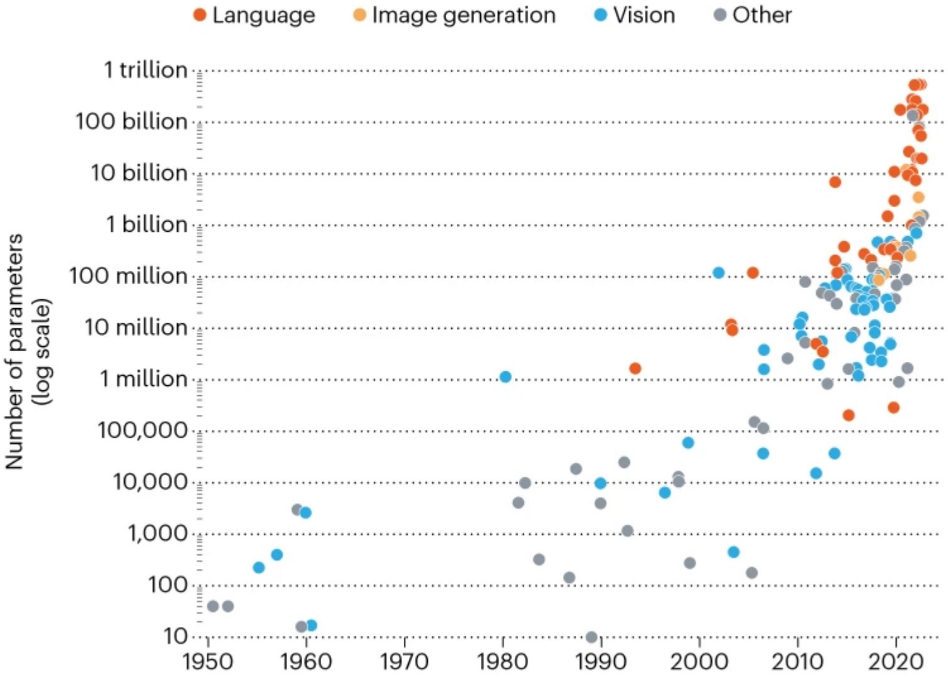


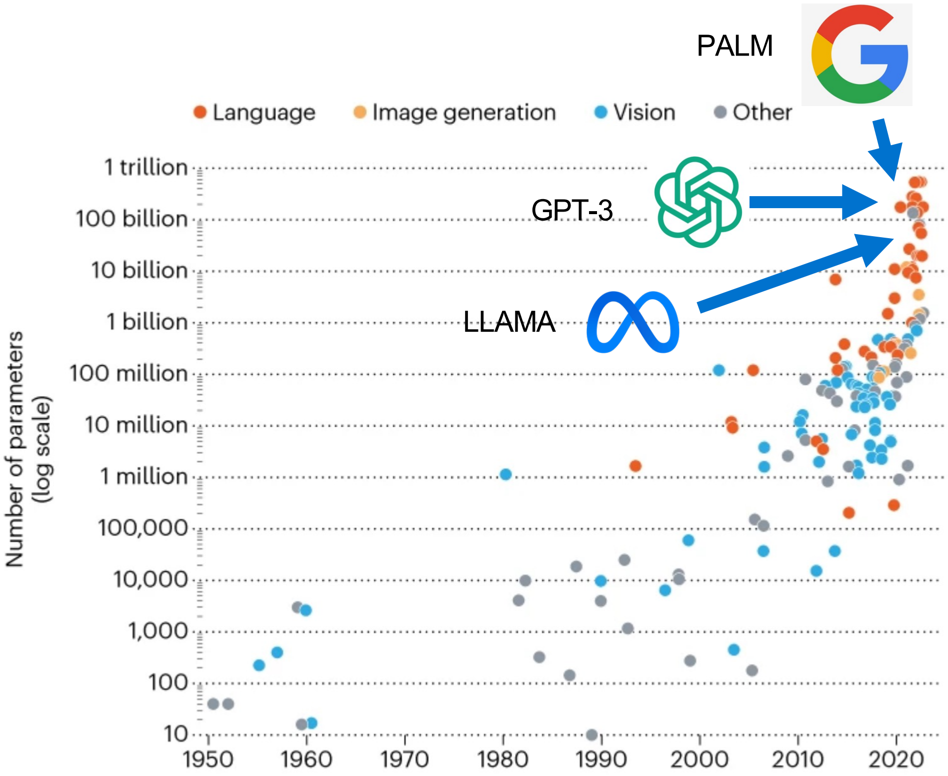
Image from: In AI, is bigger always better? https://www.nature.com/articles/d41586-023-00641-w

[Slide credit: Arman Cohan]

Current state of Self-supervised Models

State-of-the-art models are hundreds of billions of parameters

Trained on vast amounts of data (Trillions of tokens)



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[Slide credit: Arman Cohan]

On terminology

- These names are sometimes used interchangeably:
  - Self-supervised models
  - Pre-trained models
  - Generative AI models
  - Foundation models
  - Frontier models
  - …

- Though they’re not exactly the same.

On terminology

- These names are sometimes used interchangeably:
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Our models are
more than just "pretrained"!

- Though they're not exactly the same.

On terminology

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~ Models that generate content

- Though they're not exactly the same.

On terminology

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  - …

~ can be used as a foundational component of modern AI systems

That doesn't mean that these models are the foundation of AI!

- Though they're not exactly the same.

More discussion on naming (Bommasani et al., 2021) https://arxiv.org/pdf/2108.07258.pdf

On terminology

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  - …

They're user-facing "frontiers" of applications

- Though they're not exactly the same.

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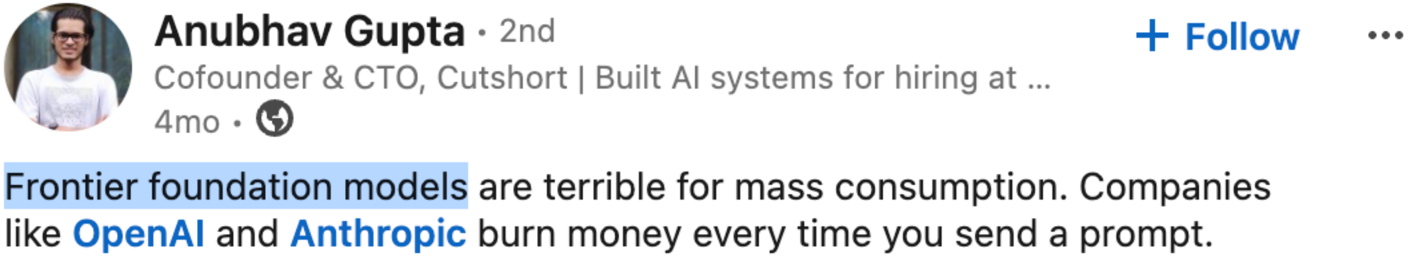
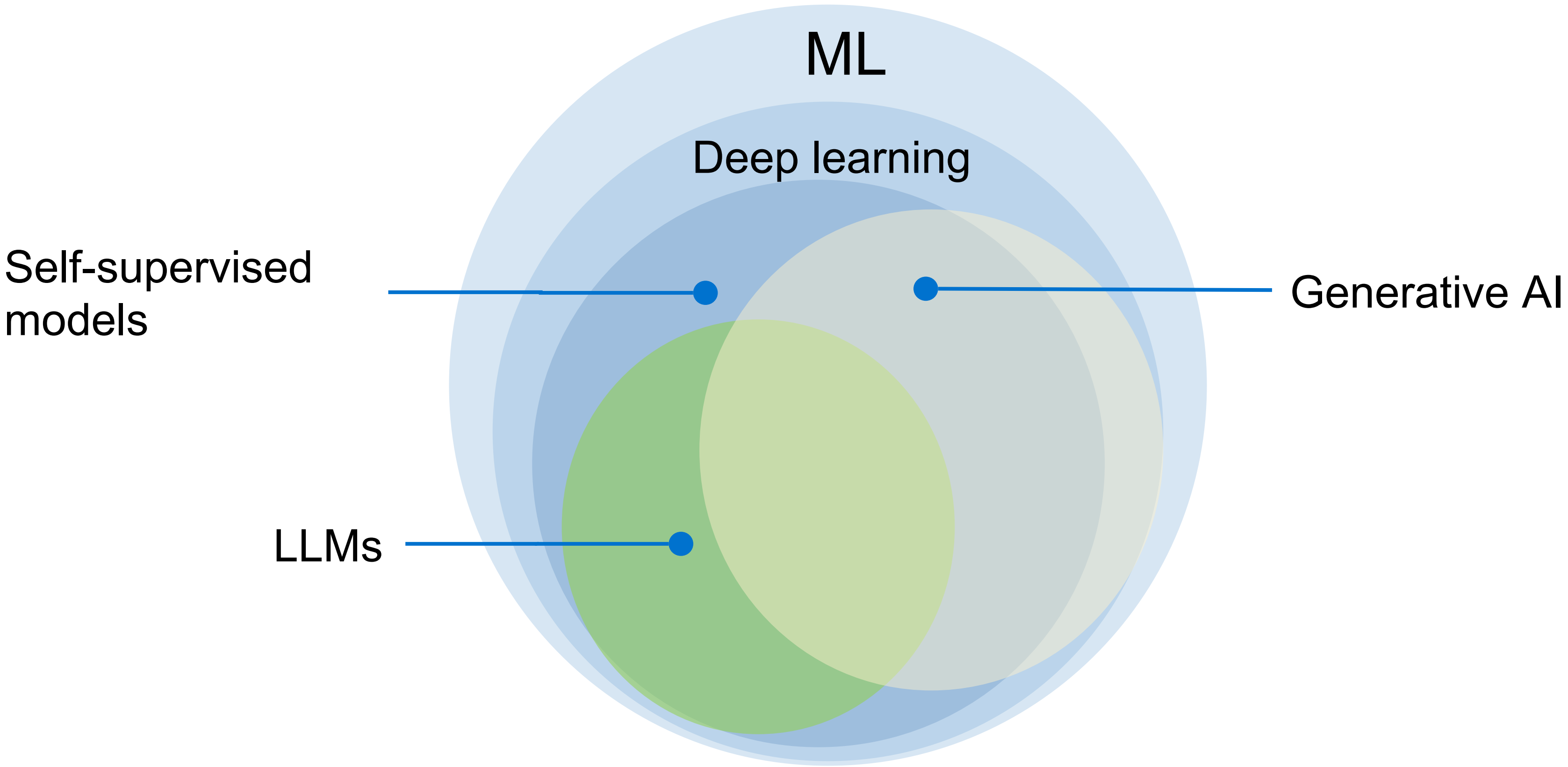
  - …

Which is your favorite? 😊

On terminology

- Also, you probably want to avoid made-up combinations
  - “frontier foundation models”!





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  - Interaction with the world (code, physical world, etc.)
  - Impacts on humans -- their misuse, biases, etc.
- Skills:
  - **Technical**—understanding of the algorithms and implementing them.
    - Gaining intuitions about **capabilities** and **limitations** of models.
  - **Soft skills**—algorithms, PyTorch, SLURM, intuition about capabilities, teamwork.

Course Logistics

CSCI 601-471/671 (NLP: Self-Supervised Models)

Course Logistics Brief

- **Instructor:** Daniel Khashabi
  - You can call me “Daniel”, as long as we act mutually respectfully.
  - No "prof Daniel"!! 🙅‍♂️

- **TA:** Jack Zhang
- **CAs:** Arda, Alvin, Lena, Matt.

Course Website

- Lots of important information on the **website**:
  - https://self-supervised.cs.jhu.edu/fa2026

Grading Policy

- In-class quizzes (individual): 70%
  - These will be ~20mins quizzes at the end of classes
  - Approximately every two weeks.
- Attendance: 0% (you can skip the lectures, if you already know them!!)
- HW: 0% (no homework assignments!!)
- Final project (team): 30% (no final exam!!)

“How should I prepare for quizzes?”

- The main source of quizzes will be the lecture slides. Make sure to understand them deeply.
- Besides the slides, you also need to spend time on “additional readings” listed on the course page.
- You get to decide how you organize your time at home.

Skip days

- Skip days: You may skip up to 2 quizzes.
- We will compute your (quiz) grade via best-of-(n-2) grades (n total quizzes)
- So, score 100% on n-2 quizzes but skip 2, your quiz grade will be 100%.
- Beyond the two skips, you will lose grade.
- The purpose is to give you flexibility for when an **unexpected** issue comes up.
- So, I suggest not wasting them too early without any purpose.

Final Project

- Must be exploring a topic related to the focus of the class.
- This is your chance to gain research experience on a topic of interest.
- Topic choice will be (relatively) free. We will help you develop your ideas!
- **Deliverables:**
  1. Submit project **proposal** outline (for our formal review and suggestions)
     - To make sure that the project is scoped reasonably and doable in your limited time.
  2. Get excited 🤩 and work on the project
  3. Final poster session + report

Communication Mechanism

- Piazza for QA
- Gradescope for your grades
- Office hours

Quick pulse check (1)

- I have understood the course expectations!
  - Yes
  - No

Quick pulse check (2)

- I am on the waitlist. Will I be able to register?
  - I don’t know.
  - However, typically, 20-30% of students drop the course within the first week. If you’re among the top 20 on the waitlist, there’s a good chance you’ll get in.

Quick pulse check (3)

- I am currently on the waitlist and very eager to take this course.
- I'm optimistic about getting in.
- Should I attend the class and take the quizzes, even though I'm not officially registered?

  - **Yes!**
  - If you're on the waitlist and excited to join, you should participate in the class and take the quizzes.
  - Everyone must adhere to the same deadlines.
  - No additional late exams for late joiners.

Wrapping it up!

- Go check the course website!
- If you’re not going to take this, drop the course!